

Data Model Documentation

Source Data

The original dataset was provided as a flat transactional table containing:

- Transaction Information
- Product Details
- Store Information
- Date & Time Attributes
- Sales Metrics

To improve reporting performance and analytical flexibility, the dataset was transformed into a dimensional model using Power Query.

Model Design

Fact Table

Sales

Contains transactional sales records and business measures.

Key Measures:

- Sales Revenue
 - Transaction Quantity
 - Transaction Count
-

Dimension Tables

Store

Store information and location attributes.

Calendar

Date hierarchy used for time intelligence calculations.

Time

Hour-level granularity used for peak-period analysis.

Product SKU

Individual product details.

Product Subcategory

Intermediate product grouping.

Product Category

High-level product classification.

Modeling Benefits

The dimensional model enables:

- Faster aggregations
- Time intelligence calculations
- Product hierarchy analysis
- Store-level performance analysis
- Scalable dashboard design

This structure follows a star-schema approach commonly used in business intelligence and reporting solutions.